

① Sample space: The set of all possible outcomes of an experiment

Ex: ① Toss a coin
↓

$S_1 = \{\text{Head}, \text{Tail}\} \rightarrow \text{sample space}$

② Tossing a coin for two times

$S_2 = \{\underline{HH}, \underline{HT}, \underline{TH}, TT\} \rightarrow \text{sample space}$

X : NO of head

$X = 2$ (for HH)
 $X = 1$ (for HT, TH)
 $X = 0$ (for TT)

$X = \{0, 1, 2\}$ ✓ Random variable

② Random variable: It is a funct^n from sample space to real number.

$X : S \rightarrow \mathbb{R}$

↓ ↓ ↓

funct^n sample space Real Num

Binomial distributⁿ: Let us have any trial with two possible outcomes $\rightarrow$ success, failure

✓ prob. of success is p (assume)

✓ prob. of failure = $1-p$

(independent)

Now we perform same trial $\downarrow$ for n time under identical condition.

Then we are assuming

X : No of success out of n trial

$\downarrow$ Then we can model this distributⁿ as binomial distributⁿ

$\downarrow$
It gives the probability of getting k^{th} success out of n trial

$$P_X(k) = {}^n C_k p^k (1-p)^{n-k}$$

such that $k = 0, 1, \dots, n$.

$${}^n C_k = \frac{n!}{(n-k)! k!}$$

Ex: We are tossing a coin for $n=10$ times and we want to find out the probability of getting 5 success out of 10 trial

$$\underline{p=0.5}$$

$$\begin{aligned} P_X(5) &= {}^{10}C_5 (0.5)^5 (1-0.5)^{10-5} \\ &= {}^{10}C_5 (0.5)^5 (0.5)^5 \\ &= \frac{10!}{5! 5!} (0.5)^{10} = 0.246 \end{aligned}$$

$$X_{\text{binom}} = [0, 1, 2, \dots, n]$$

$\nwarrow$ np.arange(0, n+1)

$$\text{PMF} = [\text{PMF}[0], \text{PMF}[1], \dots, \text{PMF}[n]]$$

$$\text{CDF} = [\text{PMF}[0], \text{PMF}[0] + \text{PMF}[1],$$

$$\text{PMF}[0] + \text{PMF}[1] + \text{PMF}[2], \dots,$$

$$\text{PMF}[0] + \text{PMF}[1] + \text{PMF}[2] + \dots + \text{PMF}[n]]$$

$$\text{CDF}[2] = {}^nC_0 p^0 (1-p)^{n-0} + {}^nC_1 p^1 (1-p)^{n-1} + {}^nC_2 p^2 (1-p)^{n-2}$$